

|               |     |                                                                                                                                  |      |   |                                                                                                                                                                                   |
|---------------|-----|----------------------------------------------------------------------------------------------------------------------------------|------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1             | (a) | $(0.5 \times) 9.3 \times 14.7 \times \sin 106$ or<br>$(9.3 \times \cos 16) \times 14.7$ or<br>$(9.3 \times \sin 74) \times 14.7$ |      | 2 | M1 for applying the area of a triangle formula using correct values (to find half of the area of the parallelogram) or for a correct method to find the area of the parallelogram |
|               |     |                                                                                                                                  | 131  |   | A1 awrt 131                                                                                                                                                                       |
|               | (b) | $(GE^2 =) 9.3^2 + 14.7^2 - 2 \times 9.3 \times 14.7 \times \cos 106$                                                             |      | 3 | M1 for the correct use of the cosine rule                                                                                                                                         |
|               |     | 377(9....) or 378 or 86.49 + 216.09 + 75.3... or 302.58 + 75.3....                                                               |      |   | M1 (dep on M1) for the correct order of operations                                                                                                                                |
|               |     |                                                                                                                                  | 19.4 |   | A1 for 19.4 – 19.5                                                                                                                                                                |
| Total 5 marks |     |                                                                                                                                  |      |   |                                                                                                                                                                                   |

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| 2             |  | $\sin 42 = \frac{6.5}{x}$ or $\frac{x}{\sin 90} = \frac{6.5}{\sin 42}$<br>or $\cos 48 = \frac{6.5}{x}$ [where $48 = 180 - 90 - 42$ ] |     | 3 | M1 or use of tan to find the horizontal side and then a correct first step in Pythagoras' theorem ie [base =] $\frac{6.5}{\tan 42}$ (= 7.21...) and<br>[x <sup>2</sup> =] $6.5^2 + "7.21..."^2$ |
|               |  | $[x =] \frac{6.5}{\sin 42}$ or $\frac{6.5 \sin 90}{\sin 42}$<br>or $[x =] \frac{6.5}{\cos 48}$ [where $48 = 180 - 90 - 42$ ]         |     |   | M1 or complete method using Pythagoras<br>[x =] $\sqrt{6.5^2 + "7.21..."^2}$<br><br>(If students give this statement with nothing before it they gain M2)                                       |
|               |  | Working not required, so correct answer scores full marks (unless from obvious incorrect working)                                    | 9.7 |   | A1 accept 9.7 – 9.72                                                                                                                                                                            |
| Total 3 marks |  |                                                                                                                                      |     |   |                                                                                                                                                                                                 |

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| 3             |  | [ADC =] $180 - 98$ (= 82)<br>[AC <sup>2</sup> =] $8^2 + 7.5^2 - 2 \times 8 \times 7.5 \times \cos(98)$ (= 136.95...)                                                                                                                                                                                                                                                                                         |      | 6 | M1 may be seen on diagram<br>M1 correct equation for AC or AC <sup>2</sup>                                         |
|               |  | [AC =] $\sqrt{136.95}$ or $\sqrt{64 + 56.25 + 16.7...}$ (= 11.7...) oe                                                                                                                                                                                                                                                                                                                                       |      |   | M1 complete method to find AC showing correct order of operations                                                  |
|               |  | eg [AD =] $\frac{"11.7" \sin 35}{\sin "82"} (= 6.77...)$ or [DC =] $\frac{"11.7" \sin 63}{\sin "82"} (= 10.5...)$ oe<br>(where "82" = $180 - 98$ , "63" = $180 - "82" - 35$ )                                                                                                                                                                                                                                |      |   | M1 correct calculation for AD or DC dep on 1 <sup>st</sup> M1 and 2 <sup>nd</sup> M1                               |
|               |  | eg [AD =] $\frac{"11.7" \sin 35}{\sin "82"}$ and [DC =] $\frac{"11.7" \sin 63}{\sin "82"}$ oe or<br>[AD =] $\frac{"11.7" \sin 35}{\sin "82"}$ and [DC =] $\sqrt{"11.7"^2 + "6.77"^2 - 2 \times "11.7" \times "6.77" \times \cos "63"}$<br>[DC =] $\frac{"11.7" \sin 63}{\sin "82"}$ and [AD =] $\sqrt{"11.7"^2 + "10.5"^2 - 2 \times "11.7" \times "10.5" \times \cos 35}$<br>Where "63" = $180 - "82" - 35$ |      |   | M1 correct calculations for AD and DC (AD = 6.77... DC = 10.5...) dep on 1 <sup>st</sup> M1 and 2 <sup>nd</sup> M1 |
|               |  | Working not required, so correct answer scores full marks (unless from obvious incorrect working)                                                                                                                                                                                                                                                                                                            | 32.8 |   | A1 accept 32.7 – 32.9                                                                                              |
| Total 6 marks |  |                                                                                                                                                                                                                                                                                                                                                                                                              |      |   |                                                                                                                    |

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| 4             | $(BC^2 =) 150^2 + 275^2 - (2 \times 150 \times 275 \times \cos 120) (= 139\,375)$                                                                                                                                                                                                                                                                                                                                                          |     | 5 | M1 for correct substitution into the cosine rule                                                                     |
|               | $(BC =) \sqrt{150^2 + 275^2 + 41250}$ oe or $\sqrt{139375}$<br>or $25\sqrt{223}$ or 373....                                                                                                                                                                                                                                                                                                                                                |     |   | M1 for correct order of operations and square root                                                                   |
|               | e.g. $\frac{\sin ABC}{275} = \frac{\sin 120}{373...}$<br>or $275^2 = 150^2 + 373...^2 - (2 \times 150 \times 373... \times \cos ABC)$<br>or $\cos ABC = \frac{150^2 + 373...^2 - 275^2}{2 \times 150 \times 373...}$<br>or $\frac{\sin ACB}{150} = \frac{\sin 120}{373...}$<br>or $150^2 = 275^2 + 373...^2 - (2 \times 275 \times 373... \times \cos ACB)$<br>or $\cos ACB = \frac{275^2 + 373...^2 - 150^2}{2 \times 275 \times 373...}$ |     |   | M1 (dep on 1 <sup>st</sup> M1) ft 373...<br><br>for a correct trig statement involving angle $ABC$<br>or angle $ACB$ |
|               | $(ABC =) \sin^{-1} \left( \frac{\sin 120}{373...} \times 275 \right) (= 39.6...)$<br>or $(ABC =) \cos^{-1} \left( \frac{150^2 + 373...^2 - 275^2}{2 \times 150 \times 373...} \right) (= 39.6...)$<br>or $(ACB =) \sin^{-1} \left( \frac{\sin 120}{373...} \times 150 \right) (= 20.3...)$<br>or $(ACB =) \cos^{-1} \left( \frac{275^2 + 373...^2 - 150^2}{2 \times 275 \times 373...} \right) (= 20.3...)$                                |     |   | M1 for a complete method to find angle $ABC$ or angle $ACB$                                                          |
|               |                                                                                                                                                                                                                                                                                                                                                                                                                                            | 140 |   | A1 accept 140 – 140.4                                                                                                |
| Total 5 marks |                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |   |                                                                                                                      |

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| 5             | eg $(AD =) \sqrt{6^2 + 6^2 - 2 \times 6 \times 6 \times \cos(50)} (= 5.07...)$ or $2 \times 6 \sin 25 (= 5.07...)$ or $\frac{6 \sin 50}{\sin 65} (= 5.07...)$ oe |      | 6 | M1 Correct expression for $AD$<br>ie $AD = \dots$ or $x =$ oe                                         |
|               | eg $6 + 6 + \sqrt{6^2 + 6^2 - 2 \times 6 \times 6 \times \cos(50)}$ or $12 + "5.07..."$<br>(=17.0)7... or 17.1)                                                  |      |   | M1 A <b>correct</b> statement of perimeter of triangle $OAD$                                          |
|               | eg $(\text{arc } BC =) \frac{50}{360} \times \pi \times 2 \times (6 + x)$ oe                                                                                     |      |   | M1 A correct statement for arc $BC$<br>(condone missing brackets around $(6 + x)$ for this mark only) |
|               | eg $2 \times "17.1" = 12 + 2x + \frac{50}{360} \times \pi \times 2 \times (6 + x)$ oe                                                                            |      |   | M1 dep on M3 for a correct equation for $x$                                                           |
|               | eg $2 \times 17.1 - 12 - \frac{30}{18} \pi = 2x + \frac{5x}{18} \pi$                                                                                             |      |   | M1 isolating terms in $x$ in a correct equation                                                       |
|               |                                                                                                                                                                  | 5.89 |   | A1 5.88 – 5.89                                                                                        |
| Total 6 marks |                                                                                                                                                                  |      |   |                                                                                                       |

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| 6             | $(AC^2 =) 9.7^2 + 12.3^2 - 2 \times 9.7 \times 12.3 \times \cos 115$<br>$(AC^2 =) 346(.2...)$ or<br>$(AC =) \sqrt{346(.2...)} \text{ or } 18.6...$                                                                                                                  |      | 5 | M1 for the correct use of cosine rule<br><br>A1 for 346 or $\sqrt{346(.2...)} \text{ or } 18.6...$                                                                         |
|               | $\frac{\sin x}{9.7} = \frac{\sin 115}{\sqrt{346}}$ oe or<br>$9.7^2 = \sqrt{346}^2 + 12.3^2 - 2 \times \sqrt{346} \times 12.3 \times \cos x$ or<br>$\frac{1}{2} \times 9.7 \times 12.3 \times \sin 115 = \frac{1}{2} \times 12.3 \times \sqrt{346} \times \sin x$ oe |      |   | M1 use of their $AC$ dep on first M1 for correct use of sine rule or cosine rule<br>or<br>for setting up an equation using the area of a triangle formula to find $\sin x$ |
|               | $\sin x = 9.7 \times \frac{\sin 115}{\sqrt{346}}$ oe or $\sin x = 0.47... \text{ or }$<br>$\cos x = \frac{\sqrt{346}^2 + 12.3^2 - 9.7^2}{2 \times \sqrt{346} \times 12.3}$ or $\cos x = 0.88...$                                                                    |      |   | M1 use of their $AC$ dep on first M1<br>Allow $(x =) \sin^{-1}(...)$ or $(x =) \cos^{-1}(...)$                                                                             |
|               |                                                                                                                                                                                                                                                                     | 28.2 |   | A1 awrt                                                                                                                                                                    |
| Total 5 marks |                                                                                                                                                                                                                                                                     |      |   |                                                                                                                                                                            |

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| 7 | $\frac{1}{2} \times 45 \times 36 \times \sin C' (= 405)$                                                                                                                                                                                                                                                                                   | <b>alternative</b><br>$\frac{2 \times 405}{36} (= 22.5)$ or $\frac{2 \times 405}{45} (= 18)$                      | 5    | M1 correct substitution into the sine area formula, with their choice of symbol to represent C.<br>or work out the perpendicular height with BC or CD as the base. |
|   | $\sin C' = \left\{ \frac{405 \times 2}{45 \times 36} \right\} (C' = 30) \text{ oe}$                                                                                                                                                                                                                                                        | $\sqrt{45^2 - 22.5^2} (= \sqrt{1518.75} = 38.97)$<br>or $\sqrt{36^2 - 18^2} (= \sqrt{972} = 31.17)$               |      | M1 correct rearrangement to make sin C the subject<br>or use Pythagoras with their found perpendicular height.                                                     |
|   | $(BD =) \sqrt{45^2 + 36^2 - 2 \times 45 \times 36 \times \cos 30'}$<br>$(= \sqrt{3321 - 3240 \times \cos 30'})$<br>$(= \sqrt{515.077...} = 22.695...)$                                                                                                                                                                                     | $\sqrt{(38.97 - 36)^2 + 22.5^2} (= \sqrt{515.077...})$<br>or $\sqrt{(31.17 - 36)^2 + 18^2} (= \sqrt{515.077...})$ |      | M1 (dep on 1st M1, ft 30) correct expression for BD ft their C (must be less than 90°).<br>or use Pythagoras to find an expression for BD.                         |
|   | $\cos ABD' = \left( \frac{22.695...^2 + 19^2 - 28^2}{2 \times 22.695... \times 19} \right)$<br>leading to 'ABD' =<br><br><b>or</b><br>$(BAD =) \cos^{-1} \left( \frac{28^2 + 19^2 - 22.695...^2}{2 \times 28 \times 19} \right)$<br>$(= 53.7...)$ <b>and</b><br>$\sin ABD' = \frac{\sin 53.7' \times 28}{22.695...}$<br>leading to 'ABD' = |                                                                                                                   |      | M1 for a complete method to find angle ABD                                                                                                                         |
|   |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                   | 83.9 | A1 accept 83.85 – 83.9                                                                                                                                             |
|   |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                   |      | <b>Total 5 marks</b>                                                                                                                                               |

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| 8 | eg $\sin 65 = \frac{AB}{8.4}$ or $\frac{AB}{\sin 65} = \frac{8.4}{\sin 90}$ |      | 3 | M1 for setting up a trig equation in AB |
|   | eg $(AB =) 8.4 \sin 65$ or $(AB =) \frac{8.4 \sin 65}{\sin 90}$             |      |   | M1 for a complete method                |
|   |                                                                             | 7.61 |   | A1 accept 7.61 – 7.613                  |
|   |                                                                             |      |   | <b>Total 3 marks</b>                    |

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| 9 | $\frac{\sin Q}{4.2} = \frac{\sin 18}{1.6}$ oe or<br>$1.6^2 = 4.2^2 + RQ^2 - 2 \times 4.2 \times RQ \times \cos 18$ oe                                                                                                                                                                                              |      | 6 | M1 correct sine ratio - could be rearranged or correct substitution into the cosine rule using angle R |
|   | $\sin^{-1} \left\{ 4.2 \times \frac{\sin 18}{1.6} \right\} (= 54.2) \text{ or } \sin^{-1} (0.811...)$<br>$\frac{2 \times 4.2 \times \cos 18 \pm \sqrt{(2 \times 4.2 \times \cos 18)^2 - 4 \times 1 \times 15.08}}{2}$                                                                                              |      |   | M1                                                                                                     |
|   | $180 - "54.2" (= 125.8) \text{ or}$<br>$RQ = 3.0585... \text{ and } 4.933...$                                                                                                                                                                                                                                      |      |   | M1 This can be implied by the correct value(s) (125.8 or 3.0585...) used later                         |
|   | $(P =) 180 - "125.8" - 18 (= 36.2)$<br><b>or</b><br>$RQ = \sqrt{4.2^2 + 1.6^2 - 2 \times 4.2 \times 1.6 \times \cos "36.2"} (= 3.0585...)$<br>or 3.0585 chosen as value from cosine rule above<br><b>or</b><br>perpendicular height = $4.2 \sin "36.2" (= 2.4805...)$<br>(where "36.2" comes from correct working) |      |   | M1                                                                                                     |
|   | $(\text{Area} =) \frac{1}{2} \times 4.2 \times 1.6 \times \sin ("36.2")$<br><b>or</b><br>$(\text{Area} =) \frac{1}{2} \times 4.2 \times 3.0585... \times \sin 18$<br><b>or</b><br>$(\text{Area} =) \frac{1}{2} \times 1.6 \times 2.4805...$                                                                        |      |   | M1                                                                                                     |
|   |                                                                                                                                                                                                                                                                                                                    | 1.98 |   | A1 awrt 1.98                                                                                           |
|   |                                                                                                                                                                                                                                                                                                                    |      |   | <b>Total 6 marks</b>                                                                                   |

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| 10            | eg $(x+5)(5x-12)=x(x+12)$                                                                                                                                                                       |      | 5 | M1 | for setting up a correct equation                                                                                                                                                                                           |
|               | eg $4x^2+x-60(=0)$ oe<br>allow $4x^2+x=60$                                                                                                                                                      |      |   | A1 | for writing the correct quadratic expression in the form<br>$ax^2+bx+c(=0)$<br><br>allow $ax^2+bx=c$                                                                                                                        |
|               | eg $(4x-15)(x+4)(=0)$<br><br>or $\frac{-1 \pm \sqrt{1^2 - 4 \times 4 \times -60}}{2 \times 4}$<br><br>or $4 \left[ \left(x + \frac{1}{8}\right)^2 - \left(\frac{1}{8}\right)^2 \right] = 60$ oe |      |   | M1 | (dep on M1) for a complete method to solve <b>their</b> 3-term quadratic (allow one sign error and some simplification – allow as far as $\frac{-1 \pm \sqrt{1+960}}{8}$ )<br>Allow + instead of $\pm$ in quadratic formula |
|               | eg $(ADE) = \sin^{-1} \left( \frac{("3.75"+5)\sin(48)}{"3.75"+12} \right)$                                                                                                                      |      |   | M1 | for a complete method for ADE.<br>Allow use of $x = -4$ for this mark                                                                                                                                                       |
|               | Correct answer scores full marks (unless from obvious incorrect working)                                                                                                                        | 24.4 |   | A1 | accept 24.3 – 24.4                                                                                                                                                                                                          |
| Total 5 marks |                                                                                                                                                                                                 |      |   |    |                                                                                                                                                                                                                             |

|               |                                                                                                                                                                                                                              |      |   |    |                    |
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| 11            | $\frac{\sin ABC}{24} = \frac{\sin 64}{31}$ oe                                                                                                                                                                                |      | 5 | M1 |                    |
|               | $(ABC) = \sin^{-1} \left( \frac{24 \times \sin 64}{31} \right) (= 44. \dots)$                                                                                                                                                |      |   | M1 |                    |
|               | $180 - "44. \dots" - 64 (= 71.9 \dots)$                                                                                                                                                                                      |      |   | M1 | accept 72          |
|               | $(DE^2) = 16^2 + 19^2 - 2 \times 16 \times 19 \times \cos "71.9 \dots"$ or<br>$(DE) = \sqrt{16^2 + 19^2 - 2 \times 16 \times 19 \times \cos "71.9 \dots"}$ or<br>$(DE) = \sqrt{617 - 181.8 \dots}$ or $\sqrt{428.166 \dots}$ |      |   | M1 | for $DE^2$ or $DE$ |
|               | Correct answer scores full marks (unless from obvious incorrect working)                                                                                                                                                     | 20.7 |   | A1 | awrt 20.7          |
| Total 5 marks |                                                                                                                                                                                                                              |      |   |    |                    |

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| 12            | eg<br>$\frac{1}{2}(2x-1)(2x+1)\sin 30 = x^2 + x - 3.75$ oe                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     | 6   | M1 | for equating area of triangle with the given area                                                          |
|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     | 3.5 | A1 | for the value of x                                                                                         |
|               | $(BC^2) = "6"^{n2} + "8"^{n2} - (2 \times "6" \times "8" \times \cos 30) (= 16.8(615 \dots))$ oe<br>or $(BC) = \sqrt{"16.8 \dots"} (= 4.10(628 \dots))$                                                                                                                                                                                                                                                                                                                                                       |     |     | M1 | ft dep on M1 for a correct method to find $BC^2$ or $BC$ ( $AB = 6$ and $AC = 8$ )                         |
|               | $\frac{\sin(ABC)}{"8"} = \frac{\sin 30}{\sqrt{"16.8"}}$ oe or $\frac{\sin(BCA)}{"6"} = \frac{\sin 30}{\sqrt{"16.8"}}$ oe or<br>$"6"^{n2} = "8"^{n2} + (\sqrt{"16.8"})^2 - (2 \times "8" \times \sqrt{"16.8"} \times \cos(BCA))$ oe or<br>$"8"^{n2} = "6"^{n2} + (\sqrt{"16.8"})^2 - (2 \times "6" \times \sqrt{"16.8"} \times \cos(ABC))$ oe                                                                                                                                                                  |     |     | M1 | ft dep on previous M1 for a correct method to find angle ABC or angle BCA                                  |
|               | $(\sin ABC) = \frac{\sin 30 \times "8"}{\sqrt{"16.8"}} (= 0.974 \dots)$ oe or $ABC = 76.9 \dots$ or<br>$(\sin BCA) = \frac{\sin 30 \times "6"}{\sqrt{"16.8"}} (= 0.730 \dots)$ oe or $BCA = 46.9 \dots$ or<br>$(\cos BCA) = \frac{"8"^{n2} + (\sqrt{"16.8"})^2 - "6"^{n2}}{2 \times "8" \times (\sqrt{"16.8"})} (= 0.682 \dots)$ oe or $BCA = 46.9 \dots$ or<br>$(\cos ABC) = \frac{"6"^{n2} + (\sqrt{"16.8"})^2 - "8"^{n2}}{2 \times "6" \times (\sqrt{"16.8"})} (= -0.226 \dots)$ oe or $ABC = 103.0 \dots$ |     |     | M1 | ft dep on previous M1 for a correct rearrangement for $\sin ABC$ or $\sin BCA$ or $\cos BCA$ or $\cos ABC$ |
|               | Correct answer scores full marks (unless from obvious incorrect working)                                                                                                                                                                                                                                                                                                                                                                                                                                      | 103 |     | A1 | accept awrt 103                                                                                            |
| Total 6 marks |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |     |    |                                                                                                            |

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| 13                              | $9^2 = 11^2 + 16^2 - 2 \times 11 \times 16 \times \cos B\hat{C}A$ oe or<br>$11^2 = 9^2 + 16^2 - 2 \times 9 \times 16 \times \cos B\hat{A}C$ or<br>$16^2 = 9^2 + 11^2 - 2 \times 9 \times 11 \times \cos A\hat{B}C$ or<br>(area of $\triangle ABC = \frac{1}{2} \times 18 \times 2 \times 7 \times 9 (= 47.6235...)$ oe                                                                                                     | 5    | M1            | For a start to the correct method to find angle $B\hat{C}A$ or angle $B\hat{A}C$ or angle $A\hat{B}C$ or a fully correct method to find the area of the triangle                                                                                                                                                                         |
|                                 | $(\cos B\hat{C}A = \left( \frac{11^2 + 16^2 - 9^2}{2 \times 11 \times 16} \right) (B\hat{C}A = 32.763...) \text{ or}$<br>$(\cos B\hat{A}C = \left( \frac{9^2 + 16^2 - 11^2}{2 \times 9 \times 16} \right) (B\hat{A}C = 41.409...) \text{ or}$<br>$(\cos A\hat{B}C = \left( \frac{9^2 + 11^2 - 16^2}{2 \times 9 \times 11} \right) (A\hat{B}C = 105.826...) \text{ or}$<br>$\frac{1}{2} \times 16 \times BD = "47.6235..."$ |      | M1            | For a correct rearrangement for $\cos B\hat{C}A$ or $\cos B\hat{A}C$ or $\cos A\hat{B}C$ or a correct equation to find $BD$ (accept angles to the nearest whole number rounded or truncated as long as not from incorrect working)                                                                                                       |
|                                 | $(BD = ) 11 \sin "32.763..." (= 5.95...) \text{ oe eg}$<br>$11 \sin (180 - "41.4..." - 105.8...) (= 5.95...) \text{ or}$<br>$9 \sin "41.4..." (= 5.95...) \text{ oe or}$<br>$\frac{"47.6235..." \times 2}{16} (= 5.95...) \text{ oe or } \sqrt{11^2 - "9.25"}^2 \text{ or } \sqrt{9^2 - "6.75"}^2$<br>$11 \sin \left( \sin^{-1} \left( \frac{9 \sin "105.826..." }{16} \right) \right) (= 5.95...) \text{ oe}$             |      | M1            | For a correct calculation that will lead to the value of $BD$ "47.6235..." may also come from $0.5 \times 9 \times 11 \times \sin "105.8..."$ or $0.5 \times 9 \times 16 \times \sin "41.4..."$ or $0.5 \times 16 \times 11 \times \sin "32.7..."$ [Students may find an angle by sine rule after already finding an angle and use this] |
|                                 | $\tan F\hat{D}B = \frac{10}{"5.95..."} \text{ oe}$                                                                                                                                                                                                                                                                                                                                                                         |      | M1            | For a correct expression for the required angle (in form $\tan x = \dots$ or $\cos x = \dots$ or $\sin x = \dots$ ) oe                                                                                                                                                                                                                   |
|                                 | Correct answer scores full marks (unless from obvious incorrect working)                                                                                                                                                                                                                                                                                                                                                   | 59.2 | A1            | awrt 59.2                                                                                                                                                                                                                                                                                                                                |
| SEE OVER FOR ALTERNATIVE SCHEME |                                                                                                                                                                                                                                                                                                                                                                                                                            |      | Total 5 marks |                                                                                                                                                                                                                                                                                                                                          |

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| Angle $D\hat{B}C = 57.237...$ Angle $A\hat{B}D = 48.591...$ $AD = 6.75 \text{ m}$ $CD = 9.25 \text{ m}$ |                                                                              |      |    |                                                                                                                        |
| 13                                                                                                      | $BD^2 = 11^2 - (16 - y)^2$ and $BD^2 = 9^2 - y^2$ oe                         | 5    | M1 | For 2 different expressions in the same single variable for $BD$ or $BD^2$                                             |
|                                                                                                         | $11^2 - (16 - y)^2 = 9^2 - y^2$ ( $y = 6.75$ or $x = 9.25$ )                 |      | M1 | Equating the 2 expressions                                                                                             |
|                                                                                                         | $BD = \sqrt{9^2 - (16 - "9.25")^2}$ or $\sqrt{11^2 - "9.25"}^2$ ( $= 5.95$ ) |      | M1 | A correct calculation to find $BD$ ("9.25" or "6.75" must come from a correct method)                                  |
|                                                                                                         | $\tan F\hat{D}B = \frac{10}{"5.95..."} \text{ oe}$                           |      | M1 | For a correct expression for the required angle (in form $\tan x = \dots$ or $\cos x = \dots$ or $\sin x = \dots$ ) oe |
|                                                                                                         | Correct answer scores full marks (unless from obvious incorrect working)     | 59.2 | A1 | awrt 59.2                                                                                                              |
| Total 5 marks                                                                                           |                                                                              |      |    |                                                                                                                        |